

Lab Report: Introduction to Database Servers & Installation

Introduction:

This lab aims to introduce database servers by installing and setting up key software tools. Databases are essential for storing and managing data for applications. We installed **.NET SDK**, **SQL Server**, and **SQL Server Management Studio (SSMS)**. Each of these tools plays an important role:

- **.NET SDK**: Allows developers to build and run applications that interact with databases.
- **SQL Server**: A database server used to store and manage large datasets efficiently.
- **SSMS**: A user-friendly tool to access, control, and manage SQL Server databases.

By the end of this experiment, we successfully set up a working database environment.

Methodology:

1. Installing .NET SDK:

- Downloaded the .NET installer from the Microsoft website.
- Ran the installer and followed the setup instructions.
- Verified installation by typing `dotnet --version` in Command Prompt, which returned the version number, confirming it was installed correctly.

2. Installing SQL Server:

- Downloaded SQL Server from Microsoft's official site.
- Chose the **Basic installation** option for simplicity.
- During setup, used **Windows Authentication** to manage access.
- After installation, made sure the server was running using the **SQL Server Configuration Manager**.

3. Installing SQL Server Management Studio (SSMS):

- Downloaded the SSMS installer from Microsoft's site.
 - Followed the on-screen instructions to install it.
 - Opened SSMS and connected to the local SQL Server instance using the provided server name.
 - Tested the connection by creating a new sample database called **TestDB**.
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Results & Discussion:

- **.NET Installation:**
The installation worked, and .NET was confirmed through the command line. This will help in running programs that connect to the database.
 - **SQL Server Installation:**
SQL Server was installed without issues and is ready to manage databases.
 - **SSMS Installation:**
SSMS connected to the SQL Server successfully. We ran a sample query, and everything worked well
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Conclusion:

The installation of .NET SDK, SQL Server, and SSMS was completed successfully. These tools are now ready to be used for developing and managing database-driven applications. We gained hands-on experience with setting up a database environment, which will help in future projects where databases play a key role.

This experiment gave us practical exposure to:

- Installing software tools for database management.
- Configuring SQL Server for local access.
- Using SSMS to interact with databases.

With everything set up, future labs will focus on building, querying, and maintaining databases efficiently.